

Foundations of Support vector machine

[Support vector machine]

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The coefficients c_i can be solved for using quadratic programming, as before. Again, we can find some index i such that $0 < c_i < (2n\lambda)^{-1}$, so that $\phi(\mathbf{x}_i)$ lies on the boundary of the margin in the transformed space, and then solve

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Parameter selection The effectiveness of SVM depends on the selection of kernel, the kernel's parameters, and soft margin parameter λ . A common choice is a Gaussian kernel, which has a single parameter γ . The best combination of λ and γ is often selected by a grid search with exponentially growing sequences of λ and γ , for example, $\lambda \in \{2^{-5}, 2^{-3}, \dots, 2^{13}, 2^{15}\}$; $\gamma \in \{2^{-15}, 2^{-13}, \dots, 2^{-1}, 2^3\}$. Typically, each combination of parameter choices is checked using cross validation, and the parameters with best cross-validation accuracy are picked. Alternatively, recent work in Bayesian optimization can be used to select λ and γ , often requiring the evaluation of far fewer parameter combinations than grid search. The final model, which is used for testing and for classifying new data, is then trained on the whole training set using the selected parameters.

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From this perspective, SVM is closely related to other fundamental classification algorithms such as regularized least-squares and logistic regression. The difference between the three lies in the choice of loss function: regularized least-squares amounts to empirical risk minimization with the square-loss, $\ell_2(y, z) = (y - z)^2$; logistic regression employs the log-loss,

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This is called the dual problem. Since the dual maximization problem is a quadratic function of the c_i subject to linear constraints, it is efficiently solvable by quadratic programming algorithms. Here, the variables c_i are defined such that

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Implementation The parameters of the maximum-margin hyperplane are derived by solving the optimization. There exist several specialized algorithms for quickly solving the quadratic programming (QP) problem that arises from SVMs, mostly relying on heuristics for breaking the problem down into smaller, more manageable chunks. Another approach is to use an interior-point method that uses Newton-like iterations to find a solution of the Karush–Kuhn–Tucker conditions of the primal and dual problems. Instead of solving a sequence of broken-down problems, this approach directly solves the problem altogether. To avoid solving a linear system involving the large kernel matrix, a low-rank approximation to the matrix is often used in the kernel trick. Another common method is Platt's sequential minimal optimization (SMO) algorithm, which breaks the problem down into 2-dimensional sub-problems that are solved analytically, eliminating the need for a numerical optimization algorithm and matrix storage. This algorithm is conceptually simple, easy to implement, generally faster, and has better scaling properties for difficult SVM problems. The special case of linear support vector machines can be solved

more efficiently by the same kind of algorithms used to optimize its close cousin, logistic regression; this class of algorithms includes sub-gradient descent (e.g., PEGASOS) and coordinate descent (e.g., LIBLINEAR). LIBLINEAR has some attractive training-time properties. Each convergence iteration takes time linear in the time taken to read the train data, and the iterations also have a Q-linear convergence property, making the algorithm extremely fast. The general kernel SVMs can also be solved more efficiently using sub-gradient descent (e.g. P-packSVM), especially when parallelization is allowed. Kernel SVMs are available in many machine-learning toolkits, including LIBSVM, MATLAB, SAS, SVMlight, kernlab, scikit-learn, Shogun, Weka, Shark, JKernelMachines, OpenCV and others. Preprocessing of data (standardization) is highly recommended to enhance accuracy of classification. There are a few methods of standardization, such as min-max, normalization by decimal scaling, Z-score. Subtraction of mean and division by variance of each feature is usually used for SVM.

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where $\mathbf{w}$ is the (not necessarily normalized) normal vector to the hyperplane. This is much like Hesse normal form, except that $\mathbf{w}$ is not necessarily a unit vector. The parameter $b / \|\mathbf{w}\|$ determines the offset of the hyperplane from the origin along the normal vector $\mathbf{w}$. Warning: most of the literature on the subject defines the bias so that

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Thus, for large values of C , it will behave similar to the hard-margin SVM, if the input data are linearly classifiable, but will still learn if a classification rule is viable or not.

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The $\mathbf{w}$ and b that solve this problem determine the final classifier, $x \mapsto \text{sgn}(\mathbf{w}^T \mathbf{x} - b)$, where $\text{sgn}(\cdot)$ is the sign function. An important consequence of this geometric description is that the max-margin hyperplane is completely determined by those x_i that lie nearest to it (explained below). These x_i are called support vectors.

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In most cases, we don't know the joint distribution of X_{n+1}, y_{n+1} outright. In these cases, a common strategy is to choose the hypothesis that minimizes the empirical risk:

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Building binary classifiers that distinguish between one of the labels and the rest (one-versus-all) or between every pair of classes (one-versus-one). Classification of new instances for the one-versus-all case is done by a winner-takes-all strategy, in which the classifier with the highest-output function assigns the class (it is important that the output functions be calibrated to produce comparable scores). For the one-versus-one approach, classification is done by a max-wins voting strategy, in which every classifier assigns the instance to one of the two classes, then the vote for the assigned class is increased by one vote, and finally the class with the most votes determines the instance classification. Directed acyclic graph SVM (DAGSVM) Error-correcting output codes Crammer and Singer proposed a multiclass SVM method which casts the multiclass classification problem into a single optimization problem, rather than decomposing it into multiple binary classification problems. See also Lee, Lin and Wahba and Van den Burg and Groenen.